KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY, KUMASI

COLLEGE OF ENGINEERING

DEPARTMENT OF AGRICULTURAL & BIOSYSTEMS ENGINEERING

ENGINEERING IN SOCIETY

CENG 291

COURSE OBJECTIVES

- To aid students appreciate the purpose of engineering to solve societal problems
- To draw a link between agricultural engineering and its application to societal problems
- To professionally and ethically prepare students for their chosen engineering field.
- To develop applicable communication, reasoning and reflection skills.

COURSE OUTLINE

- Ethics and Professionalism
- Moral Reasoning and Codes of Ethics
- Moral Frameworks
- Commitment to Safety
- Workplace Responsibilities and Rights

COURSE OUTLINE

- Workplace Responsibilities and Rights
- Honesty and Integrity
- Code of Ethics and Disciplinary Procedure of Ghana Institutions of Engineers (GHIE)
- The Canons of Engineering and Code of Ethics of the Professional Disciplines

UNIT 1 ETHICS AND PROFESSIONALISM





ETHICS

- What are ***ethics***?

Moral values that are sound, actions that are morally right, policies and laws that are desirable.

- What are ***engineering ethics***?

Responsibilities and rights that ought to be endorsed by those engaged in engineering, and also of desirable ideals and personal commitments in engineering.

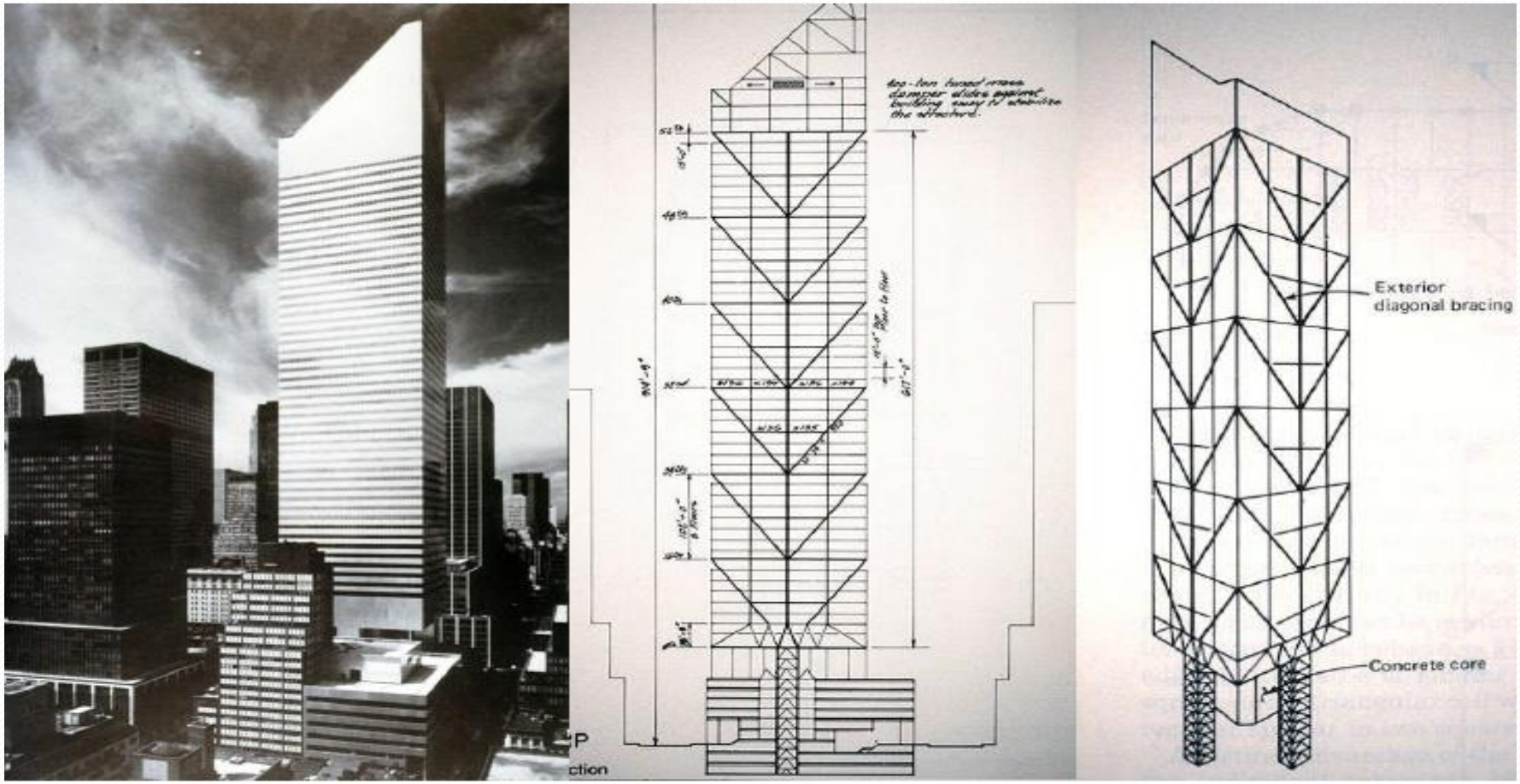
ETHICS

Why study engineering ethics?

- Contribute to safe and useful technological products
- Gives meaning to engineers' endeavours
- Strengthens one's ability to reason clearly and carefully about moral questions
- Increases **moral autonomy** (self-determining or independent)

ETHICS

Accepting and Sharing Responsibility



Engineer William LeMessurier designed the structure of the CitiCorp Tower as a braced perimeter frame with long diagonals on the facades in a chevron pattern (eight floors high) collecting and transferring the floor loads to the center of each face where the mega-columns below are located. These façade trusses collect about 1/2 of the gravity loads and resist the entire wind loads on the building. At the base of the tower, where the chevrons end, a diagonally braced transfer floor is required to transfer the wind shear (resisted by the chevron trusses) to the central concrete core.

ETHICS

❑*Accepting and sharing responsibility*

- What does it mean to be responsible?

1. Obligations (the core idea around which all other senses revolve). Obligations are role responsibilities
2. Accountable (to be answerable to an obligation)
3. Conscientious (Acceptance of obligations and diligence in fulfilling them even under difficult circumstances.)
4. Blameworthy/Praiseworthy

ETHICS

- *How to be morally autonomous or independent.*

- Moral awareness
- Cogent moral reasoning
- Moral coherence
- Moral imagination
- Moral communication
- Respect for persons
- Integrity

CODE OF ETHICS OF ENGINEERS BY ASABE

Fundamental Principles

- Engineers must use their knowledge and skill for the enhancement of human welfare
- Engineers must be honest and impartial, and serving with fidelity the public their employers and clients
- Engineers must strive to increase the competence and prestige of the engineering profession
- Engineers must support the professional and technical societies of their disciplines

CODE OF ETHICS OF ENGINEERS BY ASABE

❑ *Fundamental Canons*

Engineers shall:

- hold paramount safety, health and welfare in the performance of duties
- perform services only in the areas of competence
- issue only truthful and objective public statements
- continue their professional development throughout their careers
- provide opportunities for development of those engineers under their supervision.

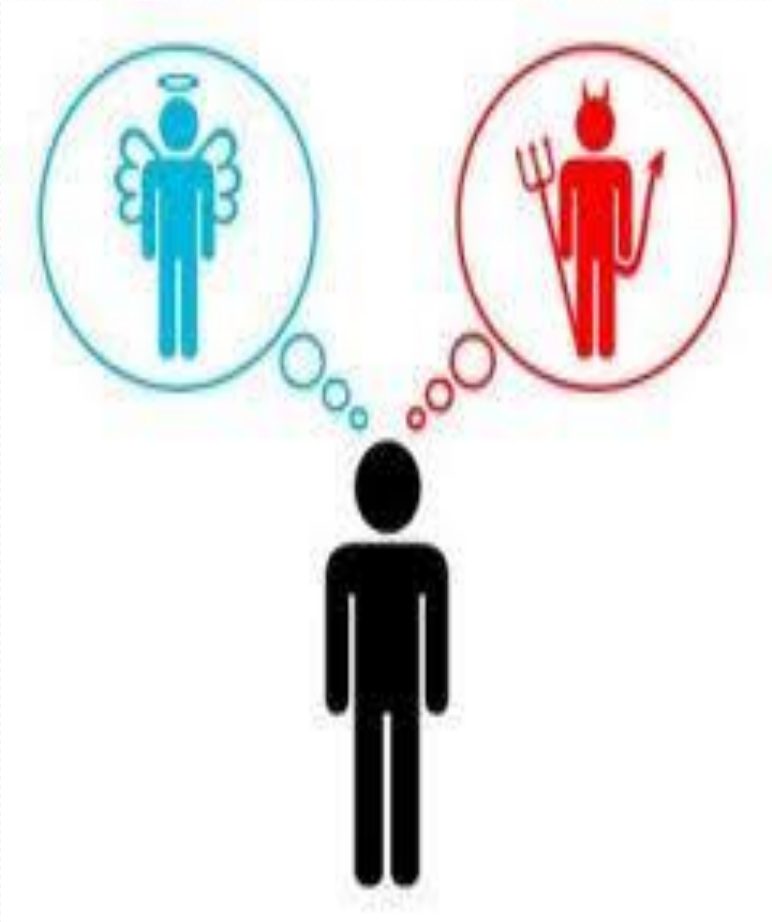
PROFESSIONALISM

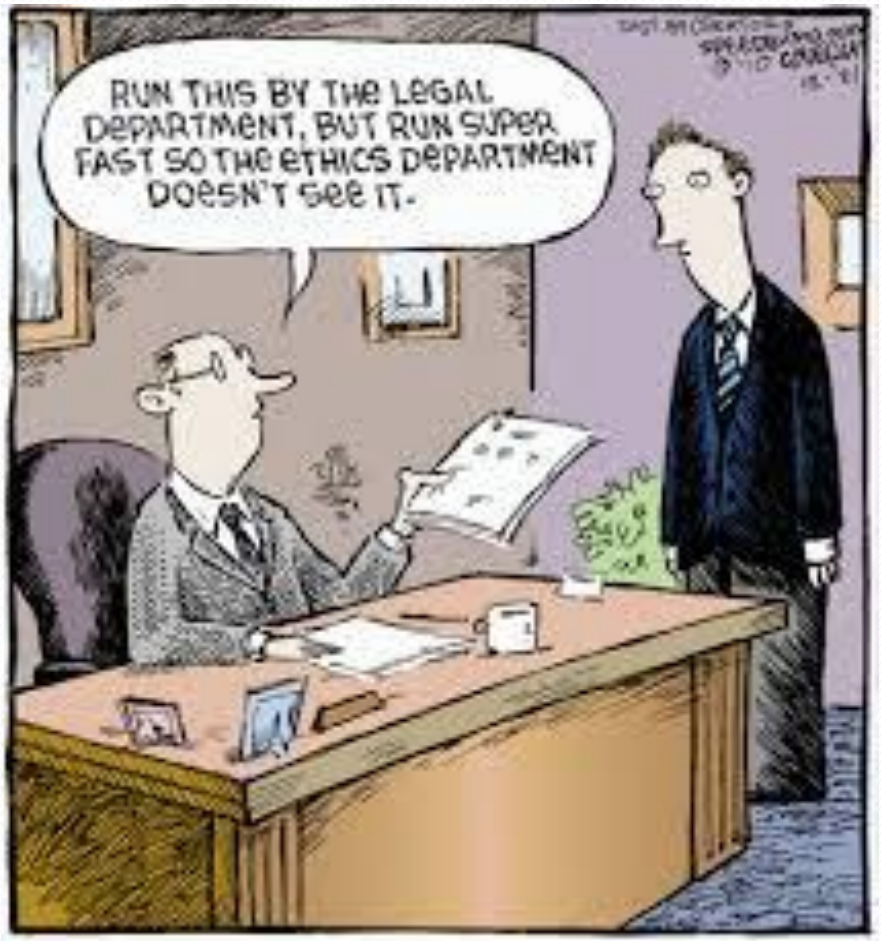
❑ *What is a **Profession**?*

Any occupation which provides a means by which to earn a living. It involves

- **Advanced expertise** (theoretical and technical know-how)
- **Self-regulation** (setting standards for admission to professions, drafting code of ethics, enforcing standard of conducts, etc.)
- Concerted service to **public good**

UNIT 2 MORAL REASONING AND CODES OF ETHICS





MORAL REASONING

❑ *Moral reasoning*

A thinking process with the objective of determining whether an idea is right or wrong. Expectedly, moral reasoning should culminate in a good decision or choice.

According to John Dewey, ***"Choice is reasonable when it...coordinates, organizes and functions each factor of the situation which gave rise to conflict, suspense and deliberation."***

MORAL REASONING

❑*Ethical (moral) Dilemmas*

Situations in which moral reasons come into conflict, or in which the applications of moral values are unclear, and not immediately obvious what should be done.

❑*Resolving Ethical Dilemmas*

The outcome of a resolved ethical dilemma is:

- Clear (moral clarity)
- Informed (Knowing and appreciating the implications of morally and relevant available facts)
- Well reasoned(good judgement on the available facts received to arrive at a morally desirable solutions).

MORAL REASONING

❑Steps in ***Resolving Ethical Dilemmas***

The following guidelines are considerable steps that will lead to a clear, informed and well reasoned choice

1. Moral Clarity: Identify the relevant moral values
2. Conceptual clarity: Clarify key concepts
3. Informed about the facts: Obtain relevant information
4. Informed about the options: Consider all options
5. Well- reasoned: Make a reasonable decision

MORAL REASONING

- Ethical Dilemmas can be divided into two main categories
  - Right-Wrong

These are dilemmas with solutions that are either right or wrong

  - Better-Worse

Dilemmas with two or more solutions, none of which is mandatory but one of which should be chosen.

- Also, the NSPE board of Ethical Review provides the valuable service to apply the NSPE code to fictionalized cases which are based on actual events. The board's rulings are available on the internet at **(http://www.niee.org)**

MORAL REASONING

Making Moral Choices

Although ethical dilemmas comprise the most difficult occasions for moral reasoning, they only constitute a small percentage of ***moral choices***. Moral choices are decisions involving moral values. They are mostly routine and straight-forward.

- The examples below entail how choices involving moral values entered into routine decisions during technological development:

1. Design of aluminium Cans
2. Design Analogy: Whitbeck

CODES OF ETHICS

Codes of ethics state the moral responsibilities of engineers as seen by the profession, and as represented by a professional society. Engineers must perform under a standard of professional behaviour that requires adherence to the highest principles of ethical conduct.

- *The **Roles** of the **Codes of Ethics** are:*

- Serving and protecting the public
- Providing guidance
- Offering inspiration
- Establishing shared standards
- Supporting responsible professionals
- Contributing to education
- Deterring wrong doing
- Strengthening a profession's image

CODES OF ETHICS

Limitation to Code of Ethics

- The may not be able to straightforwardly address all situations.
- Codes provide little guidance on the order of priority as conflict may arise.
- Codes are not always the complete and final word and as such can be flawed, both by omission and commission.
- Proliferation of codes

CODE OF ETHICS

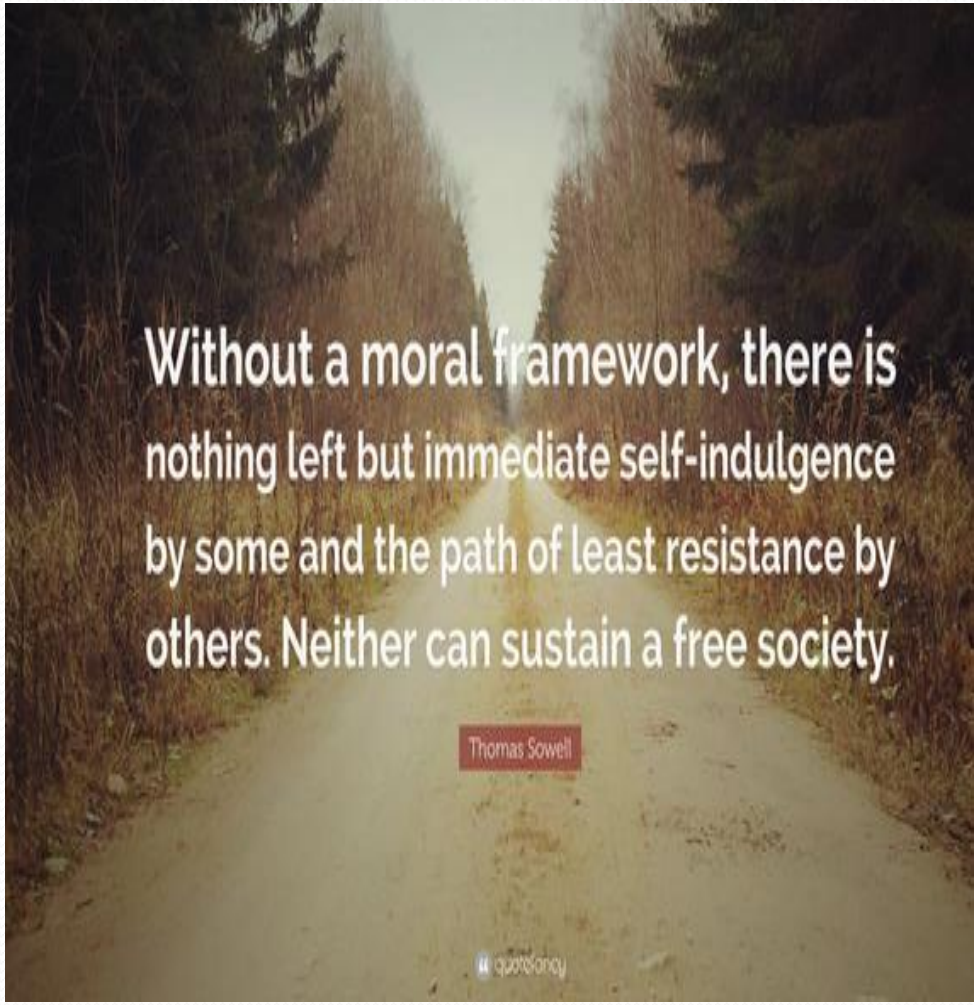
Ethical Relativism

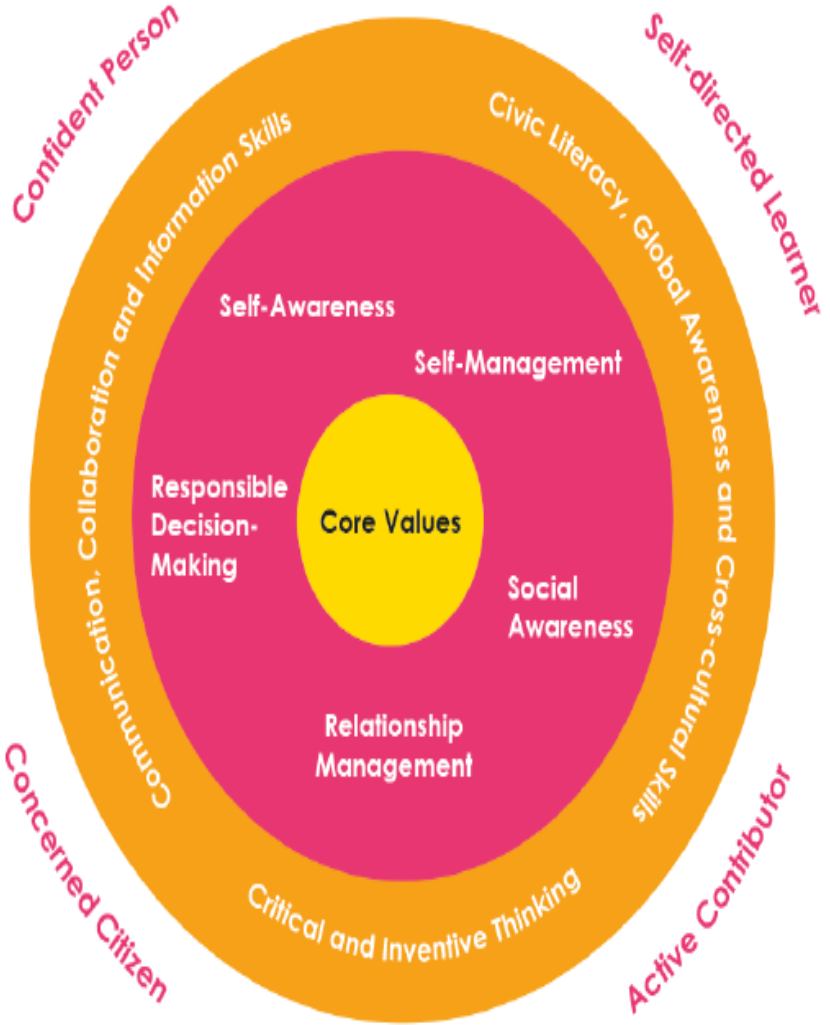
Ethical relativism is the theory that holds that morality is relative to the norms of one's culture. That is, whether an action is right or wrong depends on the moral norms of the "society" in which it is practiced.

Problems of Ethical Relativism

- Rules out the possibility of critiquing the code of ethics of another professional society.
- Gives freedom to individuals to form their own society

UNIT 3 MORAL FRAMEWORKS





MORAL FRAMEWORKS

Moral Framework

A set of ethical codes based on moral judgements that guides an individual's behaviour in choosing a best solution for a problem.

Ethical Theory

A comprehensive perspective on morality that clarifies, organises, and guides moral reflection. It grounds the requirements in engineering codes of ethics and moral values that play a role in all areas of life.

MORAL FRAMEWORK

Types of Ethical Theories

- Utilitarianism (maximise the overall good, accounting for those affected by actions)
- Right ethics (respect for human rights)
- Duty ethics (respect for individuals' autonomy)
- Virtue ethics and self realisation (good character is central to morality)
- Self realisation ethics (moral significance of self fulfilment)

MORAL FRAMEWORK

- *Utilitarianism*

The view to always produce the most good for the most people, giving equal consideration to everyone affected.

- *Cost Benefit Analysis*

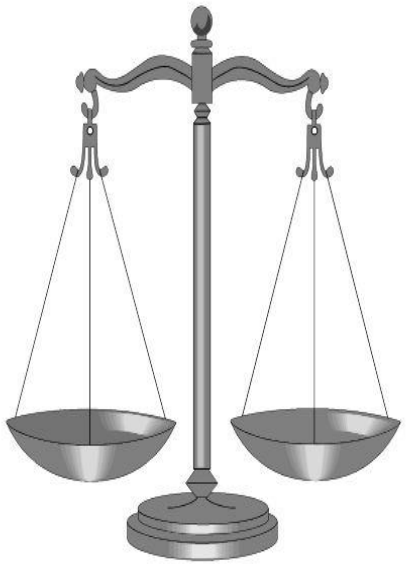
A systematic approach that estimates the strengths and weaknesses of alternatives and determine options that provide the best approach to achieve benefits while preserving savings.

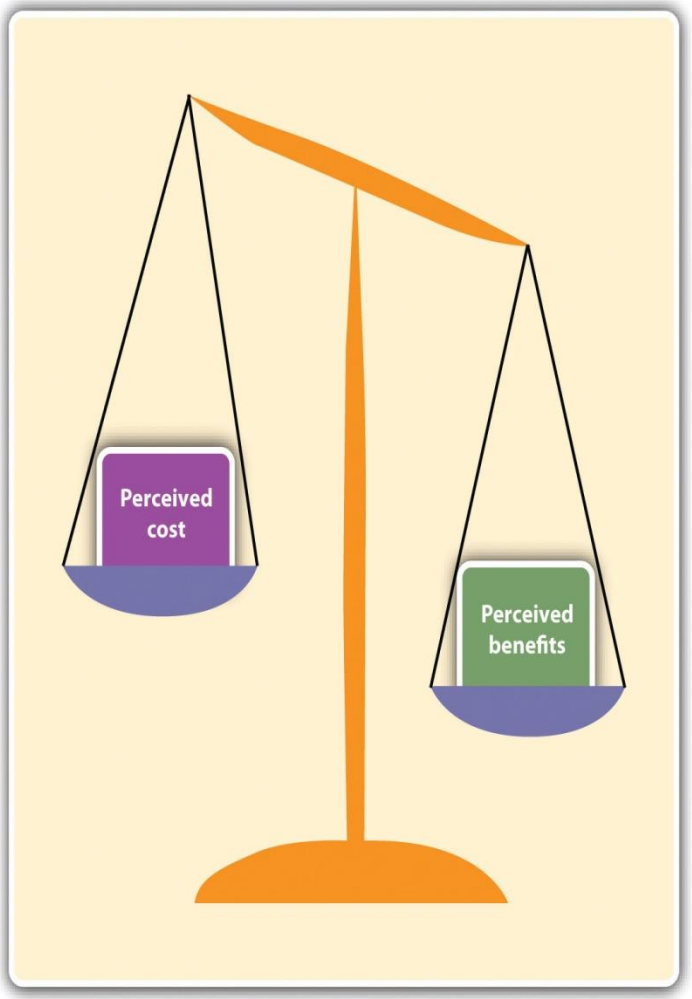
MORAL FRAMEWORK

Utilitarianism versus Cost-Benefit Analysis

The Moral Philosophy Framework

- **1) Utilitarian:**
- Consequential theories
- cost/benefit analysis
- maximize 'good' and minimize 'harm'
- Limitations:
- Measurement issues
- Whose utility curve?





MORAL FRAMEWORK

Utilitarianism versus Cost–Benefit Analysis

- Cost-Benefit analysis sounds like utilitarianism, but often it is not.
- Cost benefit analysis may usually centre around the good of a corporation, rather than the good of everyone affected, considered impartially.
- Utilitarianism consider the cost and benefits to everyone affected by a project or proposal.
- Utilitarianism weighs the interests of each person affected equally, giving no preference to members of a corporation.
- Utilitarianism adopts a long term view and do not reduce good and bad to only money.

MORAL FRAMEWORK

Act-Utilitarianism

A utilitarian theory of ethics which states that a person's act is morally right if and only if it produces the best possible results in that specific situation.

Rule-Utilitarianism

A form of utilitarian theory which states that an action is right as it conforms to a rule that leads to the greatest good.

MORAL FRAMEWORK

Act-Utilitarianism versus Rule-Utilitarianism

- Act-utilitarianism apparently permits some actions that are patently immoral but someway beneficial and causes no or little harm.
- Rule-utilitarianism demonstrate that the overall good is promoted and as such do not permit actions that are not in line with principles .
- Act-utilitarianism seems to permit injustice by promoting social good at the expense of individuals.
- Rule-utilitarianism on the other hand emphasises the general good in heeding to rules.

MORAL FRAMEWORK

- *Right Ethics*

The regard for human right as fundamental.

- *Duty ethics*

The regard for duty of respect for autonomy as fundamental.

Both theories emphasise respect for individuals' dignity and worth, in contrast with utilitarians' emphasises on general good.

MORAL FRAMEWORK

Right ethics and ***duty ethics*** are largely mirror images of each other.

That is:

Because you have a right to life, I have a duty not to kill you.

If I have a duty not to deceive you then you have a right not to be deceived.

MORAL FRAMEWORK

❑Human Rights

These constitute a moral authority to make legitimate moral demands on others to respect our choices, recognising that others can make similar claims on us.

Holding paramount the ***safety, health*** and ***welfare*** of the public can be interpreted as having respect for public's rights to ***life***, rights to ***privacy***, rights ***not to be injured***, rights to ***receive benefits through fair and honest exchanges*** in a free marketplace.

MORAL FRAMEWORK

Two versions of Right Ethics

- The first version conceives of human rights as intimately related to communities of people mainly welfare rights

- The second version of rights ethics considers only liberty rights and denies there are welfare human rights

MORAL FRAMEWORK

- Duty Ethics

These are right actions required by duties to respect the liberty or autonomy of individuals.

A writer suggests the following list of important duties

1. Don't kill
2. Don't cause pain
3. Don't disable
4. Don't deprive freedom
5. Don't deprive pressure
6. Don't deceive
7. Keep your promise
8. Don't cheat
9. Obey the law
10. Do your duty

MORAL FRAMEWORK

❑Duty Ethics

- All specific duties derive from one fundamental duty to respect persons.
- Hence, respect for persons amounts to respect for their moral autonomy.
- Duty ethics takes into accounts duty to oneself since the individual is rational and autonomous being (e.g. A duty not to kill oneself).
- Duty ethics also considers moral duties and interprets it as "categorical imperatives" (commands requiring what is right).

MORAL FRAMEWORK

Prima Facie Duties

Duties that have some permissible exceptions when they conflict with more pressing duties.

- Universality- moral rules apply to all rational agents
- Categorical imperatives- moral rules command what is right because it is right
- Absolutism-moral rules have no exceptions

MORAL FRAMEWORK

Virtue Ethics

- Virtue ethics emphasises character as the key element of ethical thinking more than rights and values.
- Virtues are the morally desirable features of an individual's character.
- Vices on the other hand are morally undesirable habits or tendencies.
- Virtues may be associated with competence, honesty, courage, fairness, loyalty and humility
- Vices may be associated with incompetence, dishonesty, cowardice, unfairness, disloyalty and arrogance.

MORAL FRAMEWORK

Virtues in Engineering

The most comprehensive virtue of engineers is responsible professionalism. This umbrella virtue implies four categories of virtues:

- Public well being(public spirited virtue focusing on the good of clients)
- Professional competence(virtues of mastery of profession and technical skills that characterises a good engineer)
- Cooperative practices(team work virtues enabling successful work with other people)
- Personal integrity(self governance virtues are those necessary in exercising responsibility).

MORAL FRAMEWORK

- *Self Realisation Ethics*

This gives greater prominence to self interest and to personal commitments that individuals develop. There are two versions to this depending on how the self is conceived.

The first, *ethical egoism*, conceived in a highly individualist manner.

In the second, the self is understood in terms of caring relationships and communities.

MORAL FRAMEWORK

Ethical Egoism

Ethical egoism says that each of us ought always and only to promote our own self interest.

The theory is *ethical* because it is a theory about morality.

It is *egoistic* because it says the sole duty of each of us is to maximise our well-being.

Psychologically we are all egoistic.

Here are two arguments to support and refute psychological egoism.

MORAL FRAMEWORK

- Argument 1

We always act on our own desires; therefore, we always and only seek something for ourselves, namely the satisfaction of our desires

In reply, the premise is true: we always act on our own desires. Actions are motivated by desires together with beliefs about to satisfy those desires. When we desire goods for ourselves, we are self-seeking; but when we desire goods for other people(for their sake), we altruistic.

The mere fact that in both instances we act on our own desires does nothing to support psychological egoism

MORAL FRAMEWORK

- Argument 2

People always ask for pleasures, therefore they always and only seek something for themselves, namely their pleasures.

In reply, there are different sources of pleasures. Taking pleasure in seeking and getting a good solely for oneself is different from taking in helping others.

MORAL FRAMEWORK

❑Motives of engineers

Engineers are motivated by self-interest, but they are also capable of responding to moral reasons in their own right as well as additional motives concerned with the particular nature of their work. Motives can be categorised into:

- Proficiency motives (centre on excellence, technical standards together with aesthetic values of beauty)
- Compensation motives (social rewards such as income, power, recognition and job or career stability)
- Moral motives (desires to meet one's responsibilities and to respect the right of others).

MORAL FRAMEWORK

What Ethical Theory is Best?

Ethical theories are attempts to provide clarity and consistency, systematic and comprehensive understanding, and helpful practical guidance in moral matters.

Sound ethical theories succeed in meeting these aims because sound ethical theories are:

- Clear and coherent
- Organises basic moral values in a systematic and comprehensive way
- Provides helpful guidance that is compatible with our most carefully considered moral convictions

COMMITMENT TO SAFETY



According to William W. Lowrance, a thing is "*safe*" if its risks are judged to be acceptable. As such safety can be judged to be relative, as can risk. For instance, a power saw in the hands of a child will never be as safe as it can be in the hands of an adult.

COMMITMENT TO SAFETY

Accidents which occur at our various engineering agencies point out how vulnerable our intricate machines and control systems can be, and how they can malfunction because of unanticipated circumstances and as such how important it is to design for proper human-machine interactions as far as safety is concerned.

Absolute safety, in the senses of

- entirely risk-free activities and products
- a degree of safety that satisfies all individuals or groups under all conditions,

is neither *attainable* nor *affordable*, yet it is important we come to some understanding of safety.

COMMITMENT TO SAFETY

Lowrance's definition of safety needs to be modified for it departs too far from our common understanding of safety. According to Lowrance, once the risks are acceptable, then it is known to be safe. Consider these situations:

- A case where we seriously underestimate the risks of something, say of buying a smartphone from a shop, only to go home and realize it was faulty and had a broken screen. On the basis of a mistaken view that buying from a shop renders the product genuine, we judge it to be safe, thus concluding we were wrong in our earlier judgement.

COMMITMENT TO SAFETY

- A case where we grossly overestimate the risks of something. For example, we irrationally think flouride in drinking water will kill a fifth of the populace. According to Lowrance, the flouridated water is unsafe, since we judge the risks to be unacceptable. It would be impossible for someone to reason with us that the water is actually safe.
- Finally, a situation in which a group makes no judgement at all about whether the risks of a thing are acceptable or not-they simply do not think about it. By Lowrance's definition, the thing is neither safe nor unsafe with respect to that group. For example, some cars are considered safe and others unsafe, yet many people drive without thinking about the safety of the cars.

COMMITMENT TO SAFETY

Because little in life, and nothing in engineering is risk-free, a modified version of Lowrance's definition is preferred.

A thing is safe if, where its risks fully known, those risks would be judged acceptable by a reasonable person in light of their settled value principles.

From this explanation, safety can now be described as how people would find risks acceptable or unacceptable if they knew the risks and were basing their judgements on their most settled value perspectives. For example, when we say an airplane travel is safer than automobile travel, we say that based on the fewer risks of death and injuries of airplane travel to automobile travel.

COMMITMENT TO SAFETY

- Risks

A risk is the potential that something unwanted and harmful may occur. A risk is taken when we undertake something or use a product that is not safe.

- Risks are somewhat unavoidable as a result of technology's influence on society.
- Some risks may be considered to be new but many have existed for some time. They are considered new only in the sense that:

1. They are now identifiable-because they have passed their accumulation threshold in our environment
2. The publics perception of them has changed-because of education, experience, media attention, etc.

COMMITMENT TO SAFETY

- Acceptability of Risk

According to William D. Rowe, a risk is acceptabe when those affected are no longer(or not) apprehensive about it. Apprehensiveness depends to a large extent on how the risk is perceived. This is influenced by factors such as:

- Whether the risk is accepted voluntarily
- The effects of knowledge on how the probabilities of harm are identified.
- Whether the effects are immediately noticeable or close at hand.
- Whether the potential victims are identifiable beforehand.

COMMITMENT TO SAFETY

- Illustration of the elements of risk perception.

- Voluntarism and Control

People are much less apprehensive about the risks to which they expose themselves voluntarily than about those to which they are exposed involuntarily. People are more willing to be subject of their own experiments than of someone else's.

- Effect of Information on Risk Assessments

The manner in which information necessary for decision making is presented can greatly influence how risks are perceived.

COMMITMENT TO SAFETY

- Job Related Risks

Exposure to risk on a job, although voluntary since one can always refuse to submit oneself to the job, yet often employees have little choice other than to do as they are told. What they are often not told is their exposure to toxic substances and dangers that are not readily sensed.

- Magnitude and Proximity

Our reaction to risk is affected by the dread of a possible mishap, in terms of magnitude and of the personal relationship we have with the victims. For example, as we feel much more keenly about a potential risk to a close friend out of a group of 20 intimate friends than to 50 strangers.

COMMITMENT TO SAFETY

Assessing and Reducing Risk

Improvement in safety of an engineered product results in an increase in the cost of the product. On the other hand, unsafe products also incur secondary costs to the manufacturer beyond primary production costs such as

- Warranty expenses
- Loss of good customer goodwill
- Loss of customers because of sustained injuries from the use of the product
- Litigation
- Possible downtime in the manufacturing process

COMMITMENT TO SAFETY

Risk-Benefit analysis

- Many large projects are justified on the basis of a risk-benefit analysis. The question answered by such are study are:

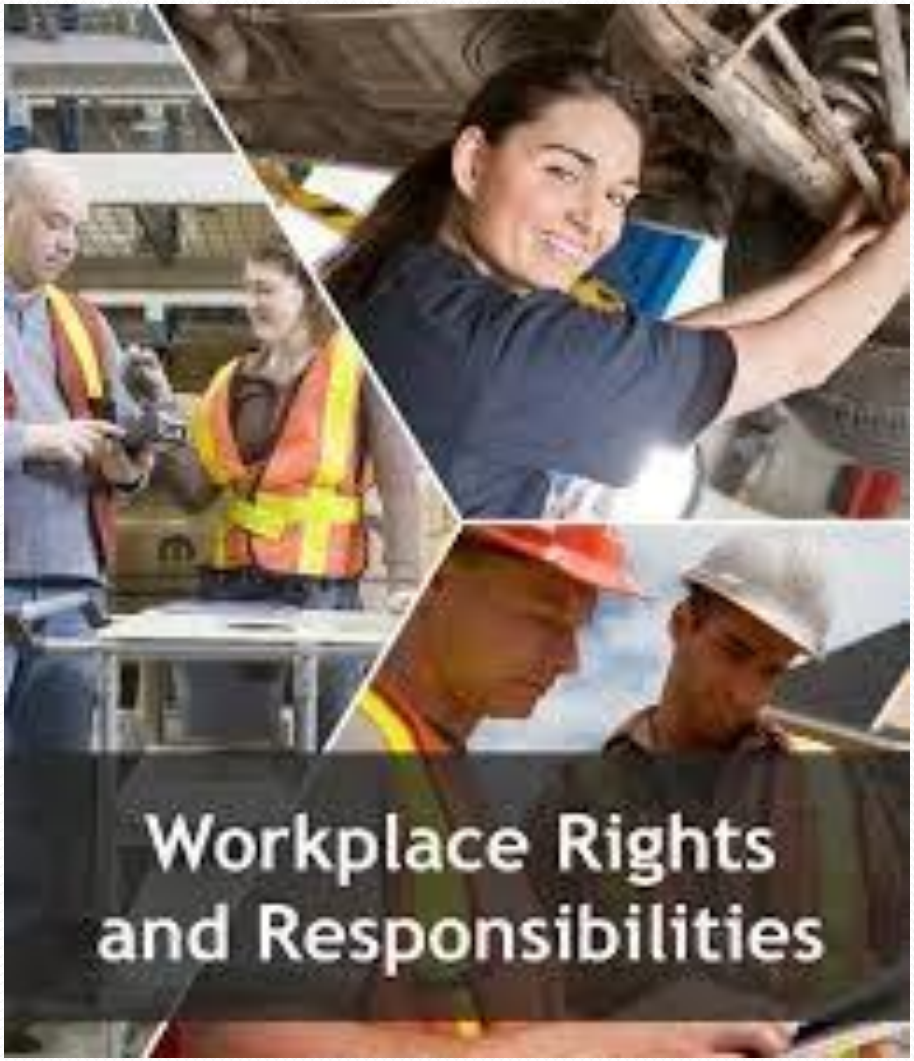
1. Is the product worth the risks connected with its use?
2. What are the benefits?
3. Do they outweigh the risks?

- If the risks and benefit can both be **readily expressed in a common set of units**, it is easy to carry out a risk-benefit analysis. However, it is more difficult to proceed when the analysis cannot be added in a common set of units, for instance assessing the effects of health, plus aesthetics plus reliability? Since these units vary from each other, it becomes relatively difficult to do a risk benefit analysis.

COMMITMENT TO SAFETY

- As such, in order to do a risk-benefit analysis, one can compare designs that satisfy some constriants and try to compare the aesthetic values with those constraints, or when the risks can be expressed and measured in one set of units, and benefits in another, we can use the ratio of risks to benefits for different designs. Like cost-benefit analysis, risk-benefit analysis is concerned with the advisability of undertaking a project.

WORKPLACE RESPONSIBILITIES AND RIGHTS





WORKPLACE RESPONSIBILITIES AND RIGHTS

Workplace

Engineers need the opportunity to perform responsibly at their workplaces thus their professional rights must be observed.

This unit looks at:

- Teamwork
- Confidentiality
- Conflicts of interest
- Professional rights
- Employee rights
- Whistleblowing

WORKPLACE RESPONSIBILITIES AND RIGHTS

❑Teamwork

The combined action of a group of people often resulting in an effective or efficient solution.

The following teamwork virtues are important in engineering:

- Loyalty to corporations
- Respect for authority
- Collegiality
- Teamwork virtues

WORKPLACE RESPONSIBILITIES AND RIGHTS

❑Ethical Corporate Climate

A working environment that is conducive to morally responsible conduct. It is produced by a combination of:

- Formal organisation and policies
- Informal traditions and practices
- Personal attitudes and commitments

WORKPLACE RESPONSIBILITIES AND RIGHTS

❑Features of an Ethical Corporate Climate

- Acknowledgement and appreciation of ethical values in in their full complexity by managers and employees alike.
- The honest use of ethical language as a legitimate part of corporate dialogue.
- Top management sets a moral tone in words, in policies and by personal example.
- Existence of a procedure for conflict resolution.

WORKPLACE RESPONSIBILITIES AND RIGHTS

❑Loyalty

Loyalty to an employer can mean two things:

- Agency-Loyalty (acting to fulfil one's contractual duties to an employer).

Agency-Loyalty is entirely a matter of actions, such as doing one's job and not stealing from one's employer regardless of the motives behind it.

- Attitude-Loyalty (deals with attitudes, emotions, and a sense of personal identity as it does with actions).

Attitude-Loyalty is often a virtue but not strictly an obligation.

WORKPLACE RESPONSIBILITIES AND RIGHTS

❑Collegiality

A kind of connectedness grounded in respect for professional expertise and in a commitment to goals and values of the profession.

The central elements of collegiality includes

- Respect for colleagues (valuing their professional expertise).
- Commitment (devotion to the moral ideals inherent in one's profession).
- Connectedness (participating in cooperative projects based on shared commitments and mutual support).

WORKPLACE RESPONSIBILITIES AND RIGHTS

Managing Conflicts

Managers have the authority and responsibility to resolve or prevent damaging conflicts that threaten corporate efficiency.

Conflict resolution often aimed at maintaining teamwork is a responsibility of all engineers.

Common conflicts confronted by engineering project managers are:

- Conflicts over schedules
- Conflicts over importance of projects and departments
- Conflicts over personnel resources
- Conflicts over technical issues
- Conflicts over administrative procedures
- Personality conflicts
- Conflicts over costs

❑Applicable Principles For Conflict Resolution

- People: Separate the people from the problem (personal aspect of conflicts is distinguished from the problem).
- Interests: Focus on interest not on positions (attention is placed on interest of personnel not position held by personnel).
- Options: Generate a variety of possibilities before deciding what to do(solutions are not compromises that split differences between stated positions).
- Criteria: Insist that the result be based on some objective standard (clear general standards should be used in evaluating results)

WORKPLACE RESPONSIBILITIES AND RIGHTS

❑Confidentiality

The duty of confidentiality is the duty to keep secret all information deemed desirable to keep secret.

Basically, it any information that the employer or client would like to have kept secret in order to compete effectively against business rivals.

Related terms to confidentiality are:

- Privileged information(available only on basis of special privilege)
- Proprietary information(trade secret, an information a company owns)
- Patents (legally protects specific products from being manufactured)

WORKPLACE RESPONSIBILITIES AND RIGHTS

❑Confidentiality and Changing Jobs

The obligation to protect confidential information does not cease when employees change jobs.

The relationship of trust between employer and employee in regard to confidentiality continues beyond the formal period of employment.

Unless the employer gives consent, former employers are barred indefinitely from revealing trade secrets.

Thus, the professional integrity of engineers involves much more than mere loyalty to one's present employer.

WORKPLACE RESPONSIBILITIES AND RIGHTS

❑Confidentiality: Justification

The primary justification is to respect the autonomy of individuals and corporations and to recognise their legitimate control over some private information concerning themselves.

Without that control, they could not maintain their privacy and protect their self interest insofar as it involves privacy.

Confidentiality has its limits, particularly when it is invoked to hide misdeeds.

Thus confidentiality will loose it bounds, when it is needed to unveil truth about white-collar crimes.

WORKPLACE RESPONSIBILITIES AND RIGHTS

Conflicts of interest

Professional conflicts of interest are situations where professionals have interest that, if pursued, might keep them from meeting their obligations to employers or clients.

Conflicts of interest can arise in the following ways:

- Gifts, bribes, and kickbacks
- Interest in other companies
- Insider information

WORKPLACE RESPONSIBILITIES AND RIGHTS

❑Gifts, bribes and kickbacks

Bribes are substantial amount of money or goods offered beyond a stated business contract with the aim of winning an advantage or keeping the contract, and where the advantage is unfair or otherwise unethical.

Gifts are not bribes as long as they are small gratuities offered in the normal conduct of business.

Kickbacks are prearranged payments made by contractors to companies or their representatives in exchange for contracts actually granted.

WORKPLACE RESPONSIBILITIES AND RIGHTS

- Interest in other companies

This consist of having an interest in a competitor's or subcontractor's business.

An example is partial ownership or substantial stockholdings in the competitor's business.

A conflict of interest will arise when one's job involves granting contracts to that subcontractor.

Moonlighting (working in one's spare time for another company) creates conflict of interest only in special circumstances such as working for competitors, suppliers, or customers.

WORKPLACE RESPONSIBILITIES AND RIGHTS

- Insider Information

A sensitive conflict of interest consisting in using “inside” information to gain an advantage or set up a business opportunity for oneself, one’s family, or one’s friends.

The information might concern one’s own company or another company which one does business.

Owning stock in the company for which ones works is not objectionable, and this is often encouraged by employers.

WORKPLACE RESPONSIBILITIES AND RIGHTS

Rights of Engineers

As employees, engineers have rights including:

- Right to receive one's salary
- Right to engage in the non-work political activities of one's choosing without coercion from employers

As professionals, engineers have special rights that arise from their professional role and the obligation it involves. These include:

- The basic right of professional conscience
- The right of conscientious refusal
- The right of professional recognition

WORKPLACE RESPONSIBILITIES AND RIGHTS

❑Right of professional conscience

- This is a moral right to exercise professional judgement in pursuing professional responsibilities.
- Pursuing those responsibilities involves exercising both technical judgement and reasoned moral convictions.
- The right of professional conscience implies more specific rights, corresponding to specific professional obligations.
- This right has limits, and must be balanced against responsibilities to employers and colleagues.

WORKPLACE RESPONSIBILITIES AND RIGHTS

Right Of Conscientious Refusal

- The right of conscientious refusal is the right to refuse to engage in unethical behaviour and to refuse do so solely because one views it as unethical.
- This is a kind of second-order right.
- Engineers have a moral right to participate in activities that are illegal and clearly unethical
- Engineers have a limited right to turn down assignments that violate their personal consciences in matters of great important such as threats to human life.

WORKPLACE RESPONSIBILITIES AND RIGHTS

Right to Recognition

Engineers have a right to professional recognition for their work and accomplishments.

Parts of this involves fair monetary remuneration, and part nonmonetary forms of recognition.

It serves as basis for criticising the unfairness of patent arrangement that fail to give more than nominal rewards to the creative engineers who make the discoveries leading to the patents.

The right to recognition, and especially fair remuneration is not only a matter of self interest but also a matter of morality too.

WORKPLACE RESPONSIBILITIES AND RIGHTS

❑Employee Rights

These are any rights, moral or legal, that involve the status of being an employee. They include:

- Institutional rights (created by organisational policies)
- Employments agreements (such as right to be paid the salary specified in one's contract)

WORKPLACE RESPONSIBILITIES AND RIGHTS

Corporations, institutions and workplaces ought to recognise a basic set of employee rights which may include:

- Privacy (right to pursue outside activities).
- Equal opportunity: non-discrimination (the right not to be discounted because of gender, race, skin colour, age etc.).
- Equal opportunity: sexual harassment (the unwanted imposition of sexual requirements in the context of relationship of unequal power).
- Equal opportunity: affirmative action (giving preference or advantage to a member of a group, in particular, women and minorities).

WORKPLACE RESPONSIBILITIES AND RIGHTS

❑ Whistleblowing

Whistleblowing occurs when an employee or former employee conveys information about a significant moral problem to someone in a position to take action on the problem, and does so outside approved organisational channels. The definition has four main parts.

1. Disclosure
2. Topic
3. Agent
4. Recipient

WORKPLACE RESPONSIBILITIES AND RIGHTS

1. Disclosure: Information is intentionally conveyed outside approved organisational (workplace) channels or in situations where the person conveying it is under pressure from supervisors or others not to do so.
2. Topic: The information concerns what the person believes is a significant moral problem for the organisation.
3. Agent: The person disclosing the information is an employee or former employee.
4. Recipient: The information is conveyed to a person or organisation that is in a position to act on the problem.

WORKPLACE RESPONSIBILITIES AND RIGHTS

Moral Guidelines

It is permissible to whistleblow when the following conditions have been met. Under these conditions there is also an obligation to whistleblow, although the obligation is prima facie and in some situations can be overridden by other moral considerations

1. The actual or potential harm reported is serious.
2. The harm has been adequately documented.
3. The concerns have been reported to the immediate superiors.
4. After not getting satisfaction from immediate superiors, regular channels within the organisation have been used to reach up to the highest levels of management.
5. There is reasonable hope that whistleblowing can help or remedy the harm.

WORKPLACE RESPONSIBILITIES AND RIGHTS

Common-sense Procedures

It is clear that a decision to whistleblow, whether within or outside an organisation, is a serious matter that deserves careful reflection.

1. Except for extremely rare emergencies, always try working first through normal organisational channels.
2. Be prompt in expressing objections. Waiting too long may create the appearance of plotting for your advantage and seeking to embarrass a supervisor.
3. Proceed in a tactful, low-key manner. Be considerate of the feelings of others involved.
4. As much as possible, keep supervisors informed of your actions, both through informal discussion and formal memorandums

WORKPLACE RESPONSIBILITIES AND RIGHTS

- Common-sense Procedures….

5. Be accurate in your observations and claims, and keep formal records documenting relevant events.
6. Consult trusted colleagues for advice – avoid isolation.
7. Before going outside the organisation, consult the ethics committee of your professional society.
8. Consult a lawyer concerning potential legal liabilities.

WORKPLACE RESPONSIBILITIES AND RIGHTS

❑Beyond Whistleblowing

- Whistleblowing is often a practical moral necessity. It holds little promise as the best possible method for remedying problems and should be viewed as a last resort.
- The obvious way to remove the need for internal whistleblowing is for management to allow greater freedom and openness of communication within the organisation.
- External whistleblowing can be avoided by the same sorts of intra-organisational modifications.
- However, conscientious engineers sometimes find the best solution to be to resign and engage in protest.

HONESTY AND INTEGRITY





HONESTY AND INTEGRITY

❑Truthfulness

- The standard of truthfulness in engineering is very high, much higher than in everyday life.
- Truthfulness imposes an absolute prohibition on deception, and in addition it establishes a high ideal of seeking and speaking the truth.
- Truthfulness also imposes that engineers must be objective and truthful and must not engage in deception.
- It is interesting to note that all other engineering codes set forth a statement of the truthfulness rule.

HONESTY AND INTEGRITY

Trustworthiness

- Trustworthiness centres on respect for autonomy and on meeting responsibilities about trust.
- Engineering is based on exercising expertise within trust relationships in order to provide safe and useful products.
- Untrustworthiness undermine expertise by corrupting professional judgements and communications.
- Untrustworthiness also undermine the trust of the public, employers and others who must rely on engineers' expertise.

HONESTY AND INTEGRITY

Academic Integrity

- Academic dishonesty includes dishonesty among students, faculty and other members of academic institutions.
- Study has revealed that among schools lacking a strong honour code, three out of four students admitted to have engaged in academic dishonesty at least once during their college career.
- Among the school with honour code, one in two made the same admission.
- Academic dishonesty among students takes several forms.

HONESTY AND INTEGRITY

- Some forms of academic dishonesty includes:

- Cheating: intentionally violating the rules of fair play in any academic exercise for example, by using crib notes or copying from another student during a test.

- Fabrication: intentionally falsifying or inventing information, for example, faking results of an experiment.

- Plagiarism: intentionally or negligently submitting others' work as one's own , for example, by quoting the words of others without using quotation marks and citing source.

- Facilitating academic dishonesty: intentionally helping other students to engage in academic dishonesty, for example, by loaning them your work.

HONESTY AND INTEGRITY

- Misrepresentation: intentionally giving false information to an instructor, for example, lying about why one missed a test.
- Failure to contribute to a collaborative project: failing to do one's fair share on a joint project.
- Sabotage: intentionally preventing others from doing their work, for example, by disrupting lab experiment.
- Theft: stealing, for example, stealing library books or other students' property.

HONESTY AND INTEGRITY

❑Research Integrity

Research in engineering requires moral ethics, however the exact moral requirements vary somewhat according to the application guidelines and regulations, but the truthfulness responsibility applies in all settings.

Research ethics has many facet which may cover:

- Research integrity and misconduct
- Conducting and reporting experiments
- Protecting research subjects
- Giving and claiming credit
- Reporting misconduct.

HONESTY AND INTEGRITY

Giving and Claiming Credit

Researchers often compete for professional recognition because it involve winning jobs, promotions and income, however outright fraud of the following types also occur:

- Plagiarism (claiming credit for someone else's ideas or work without acknowledging it, in contexts where one is morally required to acknowledge it).
- Misrepresenting credentials (researchers forge credentials, creating widely publicised scandals).
- Misleading listing of authorship (it is unethical to omit a co-author who makes a significant contribution to a research).

HONESTY AND INTEGRITY

Reporting Misconduct

Researchers have a responsibility to report misconduct by other researchers when the misconduct is serious and when they are in a position to document it.

Yet typically there are strong pressures from supervisors, colleagues, and others not to report misconduct, and hence most instances fall into the category of whistleblowing.

Measures to protect individuals who responsibly report research misconduct must be implemented and intensified.

More needs to be done to curb the stigma of “turning in” a colleague.

HONESTY AND INTEGRITY

Consulting Engineers

Consulting engineers are engineers who work in private practice. They are compensated by fees for the services they render, not by salaries received from employers.

Because of this they tend to have greater freedom to make decisions about projects they undertake.

Yet their freedom is not absolute; they share in a same light with salaried engineers the need to earn a living.

Consulting engineers have honest roles to play in:

- Advertising
- Competitive bidding
- Contingency fees

HONESTY AND INTEGRITY

Advertising

Corporate engineers are involved in advertising because they work in product sales divisions.

Consulting engineers are directly responsible for advertising their services, even when they hire consultants to help them.

It is difficult to determine whether specific ads are deceptive or not, however it is clearly deceptive for a consulting firm to claim in a brochure that it played a major role in a well-known project when it actually played a minor role.

Consumer products advertisers, are generally allowed to suppress negative aspects of the items they are promoting and to engage in some degree of exaggeration of the positive aspects.

HONESTY AND INTEGRITY

❑Competitive Bidding

Competitive bidding is competing for jobs on the basis of submitted priced proposals.

Clients do not rely exclusively on reputation and proven qualification in choosing engineering firms for contracts but also on price attached to the proposals.

This raises the problem of how the qualifications are to be determined in an equitable way.

HONESTY AND INTEGRITY

❑Contingency Fees

Contingency fee is a fee that is payable only if the outcome is successful or the outcome saves the client money.

Consulting engineers essentially make their own arrangements about payment for their work. Naturally this calls for exercising a sense of honesty and fairness.

In contingency fee situations, the consultant's judgement may easily become biased.

For example, the prospects of winning the fee may tempt the consultant to specify inferior materials in order to cut constructing costs.

HONESTY AND INTEGRITY

❑Expert Witness and advisers

Engineers increasingly are asked to serve as consultants who provide expert testimony in adversarial or potentially adversarial contexts.

Usually engineers are hired by one adversary in the dispute , and that raises special ethical concerns about their proper roles.

These engineers should function as impartial seekers and communicators of the truth.

They become ***'hired guns'*** when they are paid to tell one side of the story.

ENGINEERS AND TECHNOLOGICAL PROGRESS





ENGINEERS AND TECHNOLOGICAL PROGRESS

❑Optimism, Pessimism, Realism

Both values and facts are involved in assessing when technological change constitutes technological progress.

Progress means advancement toward valuable goals, hopefully using permissible means.

Typically, debates between optimists and pessimists turn on more than disagreements about the facts and estimates of risk.

They involve differing judgements about moral values, especially values of social justice, human fulfilments, and respect for the environment.

ENGINEERS AND TECHNOLOGICAL PROGRESS

❑Technology: Value-Neutral or Value-Laden

According to value-neutral view, technology consists of artifacts or devices - machines, tools, structures, perhaps together with knowledge about how to make and maintain devices. As such, it is neither good nor bad, but merely a means that can be used for good or bad purposes.

According to value-laden view, technology consist of more than artifacts and knowledge, but also the organisations and general approaches that make technological development possible, and organisations and approaches are guided by values.

ENGINEERS AND TECHNOLOGICAL PROGRESS

❑Shared Responsibility and Control of Technology

Technological determinism is the thesis "that technological somehow causes all other aspects of society and culture, and hence that changes in technology dictate changes in society."

In a strong version, the thesis of technological determinism denies human choice: We are victims of technology rather than in control of it.

Technological determinism undermines shared responsibility for technological projects.

For responsibility presupposes freedom, engineers are not responsible for changes that are entirely beyond human control.

ENGINEERS AND TECHNOLOGICAL PROGRESS

❑Uncertainty, Ambiguity, and Social Experimentation

Uncertainty about general trends in technological change, as well as about specific technologies, lies at the heart of debates about technological optimism.

Although the pace of scientific and engineering advancement is breath-taking, there is a lag in moral, social and political understanding.

The model of engineering as social experimentation highlights this dimension of engineering, whatever one's specific beliefs about the relevant facts concerning technological change.

ENGINEERS AND TECHNOLOGICAL PROGRESS

❑Moral Leadership

As managers, business entrepreneurs, corporate consultants, academics, and government officials, engineers provide many forms of leadership in developing and implementing technology.

In this case technology is understood broadly to include artifacts, knowledge, organisations and approaches.

ENGINEERS AND TECHNOLOGICAL PROGRESS

❑Morally Creative Leaders

Leadership is an achievement word which indicates success in ,moving a group toward goals.

Moral leaders are individuals who direct, motivate, organise, creatively manage, or in others ways move groups toward morally valuable goals.

Moral leadership means employing morally permissible means to stimulate groups to move toward morally desirable ends.

Moral leaders are morally creative, identifying the most important values that apply in a particular situation, bringing them into focus through effective communication within groups, and forming workable commitments to implement them.

ENGINEERS AND TECHNOLOGICAL PROGRESS

❑Participation in Professional Societies

Moral Leadership within engineering is often manifested in leadership within professional societies.

Professional societies do more than promote continuing education for their members.

Professional societies provide a forum for communicating, organising and mobilising change within and by large groups.

Individuals also have spurred professional societies to foster the study of engineering ethics.

ENGINEERS AND TECHNOLOGICAL PROGRESS

❑Ideals of Voluntary Service

Voluntary Service is certainly a vital avenue for providing moral leadership within communities.

By sharp contrast, most engineering codes of ethics have been silent on this issue or taken stands that discourage volunteerism.

It is important that, morally concerned engineering profession should recognise the rights of corporations and individual engineers to voluntary engage in philanthropic engineering service.